

Quiz 4

NAME: _____

Q.1. Solve the inequality $3|3x - 1| - 2 \geq 4$ and write your answer in interval notation.

Let's try and isolate for the absolute value expression:

$$\begin{aligned} 3|3x - 1| - 2 &\geq 4 \\ \Rightarrow 3|3x - 1| &\geq 6 \\ \Rightarrow |3x - 1| &\geq 2 \end{aligned}$$

Let's recall what an inequality like the one above is saying in "distance terms". This inequality is saying that $3x - 1$ is at a distance at least 2 from the origin of the real line (i.e. from 0). So we get two cases for the quantity $3x - 1$:

$$\begin{array}{lll} 3x - 1 \geq 2 & \text{OR} & 3x - 1 \leq -2 \\ 3x \geq 3 & \text{OR} & 3x \leq -1 \\ x \geq 1 & \text{OR} & x \leq -\frac{1}{3} \end{array}$$

so our solution set is $x \geq 1$ or $x \leq -\frac{1}{3}$. In interval notation, this is $\left(-\infty, -\frac{1}{3}\right] \cup [1, \infty)$.

Q.2. Solve the inequality $\frac{8}{x+3} < 2$ and write your answer in interval notation (**Hint:** start by getting 0 on side of the inequality).

First let's state the domain of this inequality (since it involves division). We cannot let any denominator be 0, so we need to take $x \neq -3$ as our domain.

Now let's follow the hint and try and rearrange and simplify this inequality:

$$\begin{aligned} \frac{8}{x+3} &< 2 \\ \Rightarrow \frac{8}{x+3} - 2 &< 0 \\ \Rightarrow \frac{8}{x+3} - \frac{2(x+3)}{x+3} &< 0 \\ \Rightarrow \frac{2-2x}{x+3} &< 0 \end{aligned}$$

So we need to figure out when $\frac{2-2x}{x+3}$ (or $\frac{2(1-x)}{x+3}$) is negative. We can do this using a sign table. The roots of the factors in this expression are $x = 1, -3$. Deleting these roots from the real line,

we get the three intervals $(-\infty, -3)$, $(-3, 1)$, $(1, \infty)$:

	$(-\infty, -3)$	$(-3, 1)$	$(1, \infty)$
$2 - 2x$	+	+	-
$x + 3$	-	+	+
$\frac{2-2x}{x+3}$	-	+	-

We see that $\frac{2-2x}{x+3}$ is negative only in the intervals $(-\infty, -3)$ and $(1, \infty)$, so our final answer is $(-\infty, -3) \cup (1, \infty)$.

Q.3. Write the **general expression/ formula** for the midpoint of the two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. Then use this formula to find the midpoint of the points $A(-1, 7)$ and $B(5, 3)$.

The general formula of the midpoint of $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is:

$$MP = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

In this problem $P_1(x_1, y_1) = A(-1, 7)$ and $P_2(x_2, y_2) = B(5, 3)$, so $x_1 = -1, y_1 = 7$ and $x_2 = 5, y_2 = 3$, giving us a midpoint:

$$MP = \left(\frac{-1 + 5}{2}, \frac{7 + 3}{2} \right) = (2, 5)$$

so our final answer is $(2, 5)$.

Q.4. Determine with working whether or not the point $C(4, 8)$ lies on the perpendicular bisector of the line segment from $A(-1, 7)$ to $B(5, 3)$.

The perpendicular bisector of A and B is exactly the set of points equidistant to A and B . So if C is equidistant to A and B it lies on the perpendicular bisector, and if it is not equidistant to A and B , then it does not lie on the perpendicular bisector.

The general formula for the distance between $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is:

$$d(P_1, P_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Let's compute the relevant distances in this problem:

$$\begin{aligned} d(C, A) &= \sqrt{(4 - (-1))^2 + (8 - 7)^2} \\ &= \sqrt{5^2 + 1^2} \\ &= \sqrt{26} \end{aligned}$$

and:

$$\begin{aligned}d(C, B) &= \sqrt{(4 - 5)^2 + (8 - 3)^2} \\&= \sqrt{(-1)^2 + 5^2} \\&= \sqrt{26}\end{aligned}$$

so $d(C, A) = d(C, B)$, i.e. C is equidistant to both A and B , meaning that yes, C does lie on the perpendicular bisector of the line segment from A and B .